Junyang Zhang

Hefei, China zhangjunyang@mail.ustc.edu.cn

Education

University of Science and Technology of China (USTC) — *Hefei, China*
PhD. in Computer Science and Technology — *Sep 2022 – Present*
Supervised by IEEE/ACM Fellow Prof. Xiang-Yang Li

University of Science and Technology of China (USTC) — *Hefei, China*
M.S. in Computer Science and Technology — *Sep 2020 – Jul 2022*
Supervised by IEEE/ACM Fellow Prof. Xiang-Yang Li

Chongqing University (CQU) — *Chongqing, China*
B.E. in Computer Science and Technology — *Sep 2016 – Jul 2020*

Research Interests

- **Edge Intelligence**: Edge AI, Model Efficiency & Compression, Distributed Inference, Edge–cloud Collaboration, Heterogeneous Devices, Industrial edge / IIoT.
- **Intelligent Sensing**: Ubiquitous Computing; Signal Processing; Side Channel.

Publications

The following are papers I have published as the first author and co-first author.

1. A-VL: Adaptive Attention for Large Vision-Language Models. **Junyang Zhang**, Mu Yuan, Ruiguang Zhong, Puhan Luo, Huiyou Zhan, Ningkang Zhang, Chengchen Hu, Xiangyang Li. **AAAI**, 2025. [Paper] [Code] [Slides]
2. WordWhisper: Exploiting Real-Time, Hardware-Dependent IoT Communication Against Eavesdropping. **Junyang Zhang**, Jiahui Hou, Ye Tian, Xiang-Yang Li. **IEEE TMC**, 2024. [Paper]
3. TensAllo: Adaptive Deployment of LLMs on Resource-Constrained Heterogeneous Edge Devices. Bowen Zhang†, **Junyang Zhang**†, Jiahui Hou and Yixin Wang. †*Co-first authors.* **IEEE INFOCOM**, 2025. [Paper]
4. Deploy Efficient Large Language Model Distributed Inference Pipeline for Heterogeneous GPUs. **Junyang Zhang**, Jiahui Hou, Bowen Zhang and Xiang-Yang Li. **IEEE/ACM IWQoS**, 2025. [Paper]

Research Experiences

Intelligent Cockpit Inference Optimization Based on Multimodal LLMs — *USTC/NIO*
USTC-NIO Joint Laboratory Project — *Jan 2024 – Mar 2025*

- Focus: Multimodal vision–language LLMs can enhance in-cabin user experience, yet strict on-device compute and memory budgets make inference efficiency the primary bottleneck for deployment.
- Profiled cross-modal attention to identify redundant computation; designed and implemented an adaptive attention module that selectively reduces unnecessary attention states during decoding. Validated on NIO smart vehicles, achieving 49% decode latency reduction and 43% KV-cache memory reduction.
- Contribution: First author of a paper accepted at **AAAI 2025**; led core idea formulation, module implementation, experimental design, evaluation, and manuscript writing.

Parallel Inference for LLMs in Edge Weak Devices — *USTC*
Research Project — *Feb 2024 – Jul 2024*

- Focus: Enable privacy-preserving, low-latency LLM inference for smart-home assistants by distributing computation across CPU-only, heterogeneous IoT edge devices under strict memory and power constraints.
- Built a distributed edge inference framework that jointly optimizes parallel execution and quantization, and introduced a power-aware device allocation strategy to balance latency, communication overhead, and energy. Implemented and deployed on resource-constrained edge devices, achieving up to 50% latency reduction.
- Contribution: Co-First author of a paper accepted at **INFOCOM 2025**, contributed end-to-end to conceptualization, methodology design, experimental validation, and manuscript writing.

Distributed Pipeline Deployment of LLMs on Heterogeneous GPUs — *USTC*
USTC–Huawei University-Industry Collaboration Project — *Jan 2023 – Jul 2024*

- Focus: Existing LLM inference pipelines are mostly designed for homogeneous GPU clusters; on heterogeneous GPUs, stage imbalance can cause severe blocking and GPU idling (idle time can exceed 50%), becoming a key bottleneck for efficient deployment.
- Proposed a heterogeneity-aware pipeline deployment framework that models GPU capability differences and automatically generates an efficient pipeline. Evaluated on 26 heterogeneous GPUs, improving makespan by 37.1% on average over baselines.
- Contribution: First author of the paper accepted at **IWQoS 2025**; led problem formulation and overall system design, implemented the prototype, and conducted evaluation and paper writing.

ESG Domain-Specific Question Answering System *USTC*
Baidu University–Industry Collaboration Project. System Development. *Jan 2023 – Jan 2024*

- Focus: ESG knowledge is large-scale and highly heterogeneous; an expert QA system based on a RAG architecture is needed to improve information access and decision-making efficiency.
- Co-developed an industry ESG knowledge base and an LLM-powered RAG QA system. Optimized core RAG components including document parsing and retrieval/recall; fine-tuned an embedding model on ESG corpora to improve retrieval accuracy.
- Contribution: Collaborated with the R&D team to build a $300GB+$ ESG document knowledge base; delivered and launched an ESG intelligent QA system, displayed on Baidu's ESG homepage.

Side-Channel Communication for IoT Devices *USTC*
Conducted under an advisor-led MIIT IoT Security Key R&D project *Jan 2021 – Jan 2022*

- Focus: Near field communication on IoT devices is vulnerable to eavesdropping. We aim to build a hardware-dependent channel so that only the intended device can correctly receive and decode messages.
- Proposed a real-time, word-level speaker-to-gyroscope communication system that leverages hardware-dependent MEMS gyroscope resonance to make non-target devices receive ineffective information. Built a decoding model trained and updated using synthesized data from theoretical resonance analysis and handled multiple eavesdroppers.
- Contribution: First author of the paper accepted at **IEEE TMC**; led problem formulation, theoretical resonance analysis, system design and implementation, experimental evaluation, and manuscript writing.

User Activity Recognition Based on Wearable Devices *USTC*
Huami Technology University–Industry Collaboration Project *Mar 2020 – Dec 2020*

- Focus: High-accuracy detection of users' physical activities (e.g., walking, running, cycling, ball sports) is essential for wearable bands and smartwatches.
- Designed a lightweight LSTM+CNN model to jointly learn spatiotemporal features from gyroscope and accelerometer signals; collected an activity dataset and developed the end-to-end recognition pipeline.
- Contribution: Improved activity recognition accuracy, with notable gains on activities involving frequent hand movements by leveraging multi-sensor spatial features.

Academic Awards

Special Talent Cultivation Project for PhD Students, China Assoc. for Sci. & Tech. (¥40000) Dec 2025
National Scholarship (top 0.2% in China), Ministry of Education, China (¥30000) Oct 2025
Annual Graduate Academic Scholarship, USTC (¥13000-18000 per year) Sep 2020-2025
Second Prize, Ubiquitous Intelligent Sensing Technology Innovation Application Competition, CCF Nov 2024

Internships & Services

Algorithm Engineer Intern, NIO Inc. *Hefei, China*
Optimizing large vision-language model inference for intelligent cockpits. *Sep 2023 – Mar 2025*

Teaching Assistant, Freshman Seminar Sep 2024 – Present
Teaching Assistant, Applied Mathematics for Computer Science Feb. 2023 – Jul. 2023
Teaching Assistant, Intelligent Internet of Things Feb. 2021 – Jul. 2021
Volunteer, China Turing Conference Jul. 2022